

INM434 Natural Language Processing

2025-26 Coursework Video Presentation

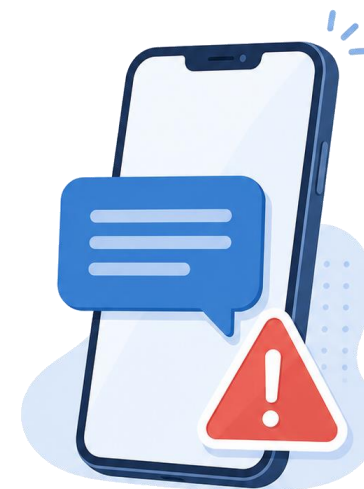
"From Bag-of-Words to Transformers: A Four-Tier Comparison of SMS Spam Detection Methods with Adversarial Robustness Analysis"

Oleh Hastov

University Id: 250064874

MSc Artificial Intelligence

Oleh.Hastov@citystgeorges.ac.uk



Problem and motivation



- SMS is still widely used for real communication and spam
- Clean spam detection showed strong results
- The harder question is robustness under text changes
- Spammers may hide spam using obfuscation
- I evaluate clean performance and robustness together



Research questions

- Which model family performs best on clean SMS spam classification?
- Do models stay robust when spam messages are obfuscated?
- Does ham side perturbation reveal hidden OOV bias?
- Can adversarial training reduce this weakness?
- Is cross seed testing needed for reliable conclusions?



Dataset and split protocol

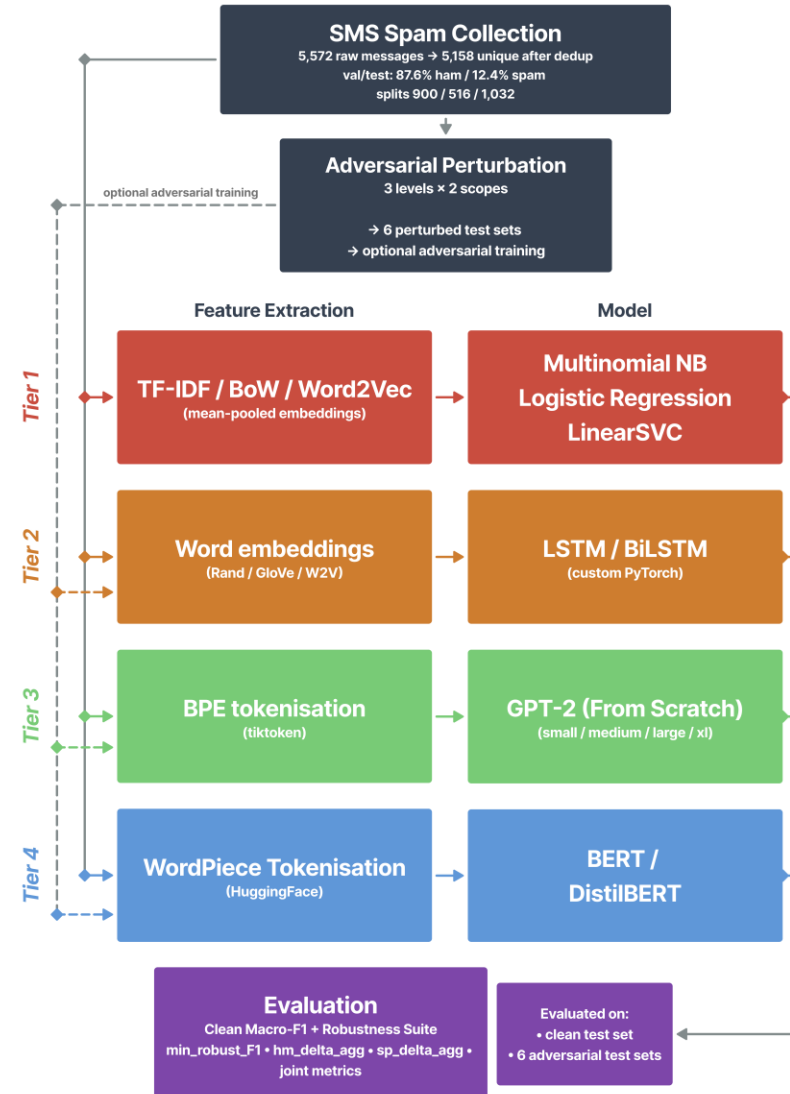


- SMS Spam Collection dataset
- 5,572 raw messages
- 5,158 unique messages after deduplication
- Training split balanced to 450 ham and 450 spam
- Validation and test keep the natural 12.4 percent spam ratio
- Clean test set: 1,032 messages

Split	N	Spam %	Distribution
Train	900	50.0%	Balanced (450 / 450)
Validation	516	12.4%	Natural
Test (clean)	1,032	12.4%	Natural
Test (adv x 6)	6,192	12.4%	Natural

Four tier NLP pipeline

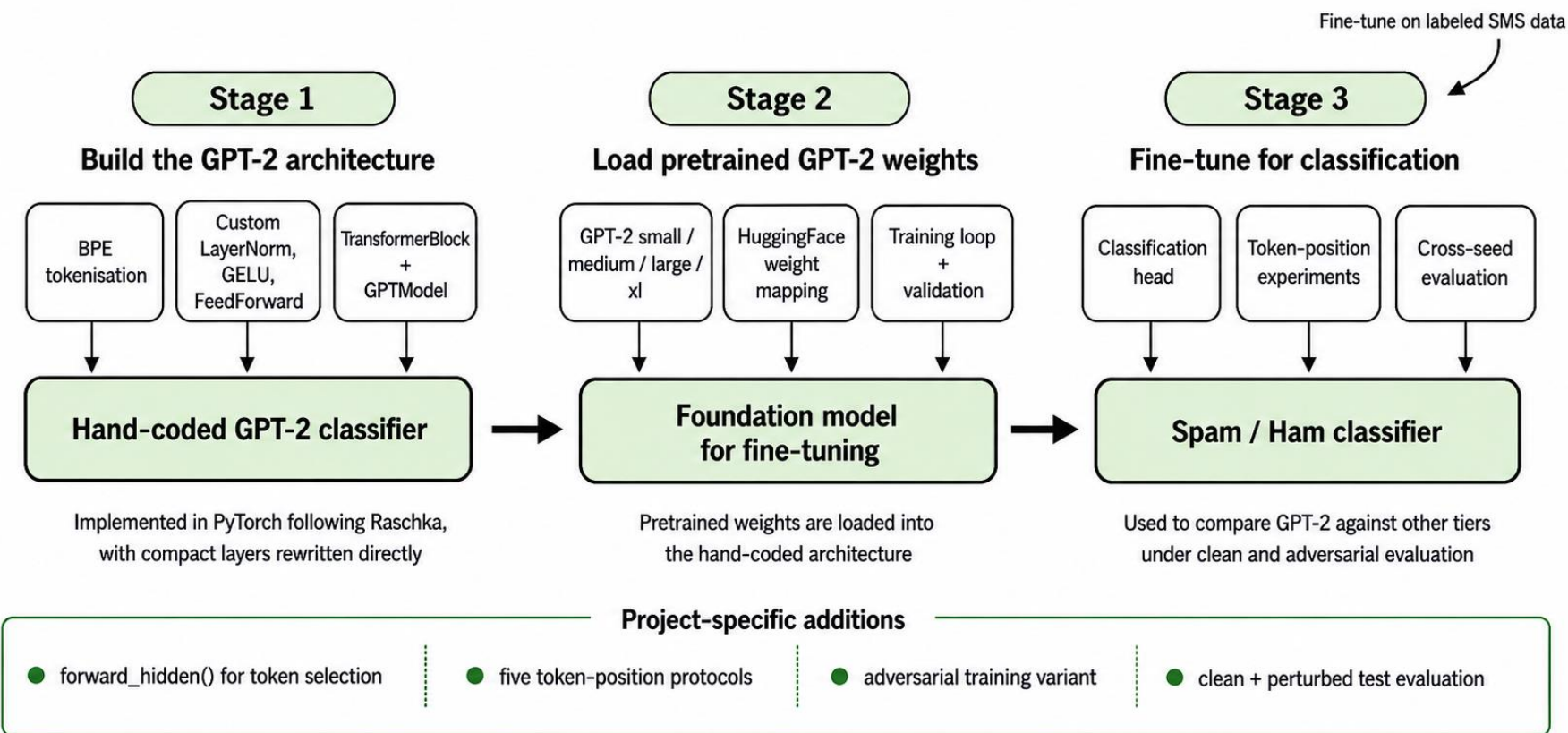
- Tier 1: sparse vectorizers with sklearn classifiers
- Tier 2: LSTM and BiLSTM models in PyTorch
- Tier 3: GPT 2 from scratch with pretrained weights
- Tier 4: BERT and DistilBERT fine tuning
- All tiers use the same splits and evaluation logic



Tier 3: GPT-2 From Scratch



Adapted for SMS spam classification in this project



- Hand-coded GPT-2 classifier following Raschka
- Pretrained GPT-2 weights loaded into my architecture
- Fine-tuned for spam and ham classification
- Added token-position tests, cross-seed evaluation and adversarial training

Evaluation methodology



- Clean macro F1 is the baseline metric
- Spam side perturbation tests realistic obfuscation
- Ham side perturbation is a diagnostic stress test
- Six adversarial test sets are used
- Cross seed testing checks stability
- Adversarial training tests mitigation

Evaluation part	Purpose
Clean test	Standard model performance
Spam side perturbation	Realistic spammer behaviour
Ham side perturbation	Diagnostic OOV bias test
Cross seed runs	Ranking reliability
Adversarial training	Mitigation check

Clean test results



- All four tiers reach high clean macro F1
- Tier 1 best joint cell: 0.939
- Tier 2 reference BiLSTM cell: 0.948
- GPT 2 small reaches 0.978
- BERT also reaches 0.978
- Clean F1 alone does not decide the best model

Tier	Best joint cell	Interpretation	F1
T1	A05	Strong cheap baseline	0.939
T2	T2_13	BiLSTM improves on sparse	0.948
T3	F_d30	GPT-2 ties BERT on clean	0.978
T4	T4_bert_lr2e-5	Strongest encoder baseline	0.978

Final selected models and parameters



Selection rule: deployment-focused joint_deploy

- I choose one best overall model and one best non-BERT model under realistic spam obfuscation.

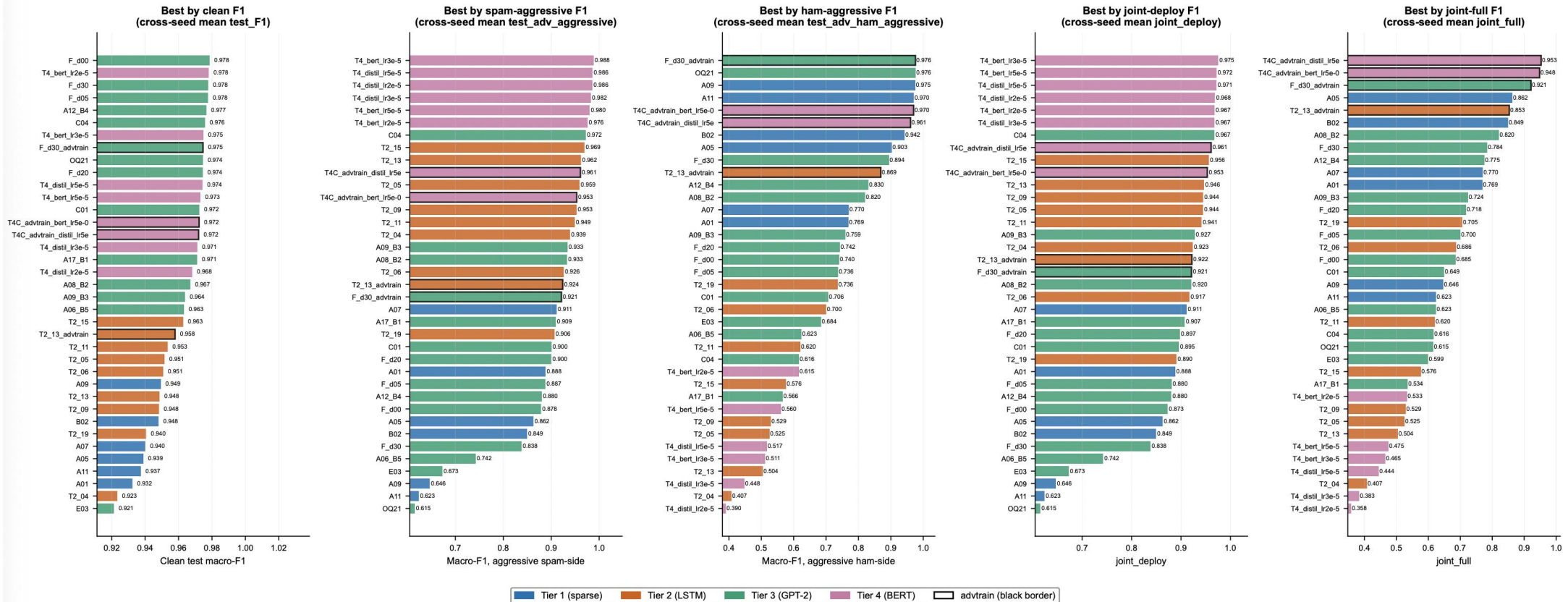
Selection role	Model	Alias	Key parameters	Why selected
Best overall deployment model	BERT-base-uncased fine-tuned classifier	T4_bert_lr3e-5	Pretrained BERT-base-uncased; WordPiece tokenisation; Max length 128; Learning rate 3e-5; [CLS] classification head; AdamW + linear warmup; 5-seed evaluation	Highest joint_deploy score overall: 0.975
Best non-BERT deployment model	Hand-coded GPT-2 small classifier	T3 C04	GPT-2 small (124M); 12 layers; 12 heads; Embedding size 768; BPE tokenisation with tiktoken; Pretrained weights loaded; Flexible_token protocol; All layers trainable; Learning rate 5e-5; 5-seed evaluation	Strongest non-BERT model by joint_deploy: 0.967

Selection criteria change the ranking

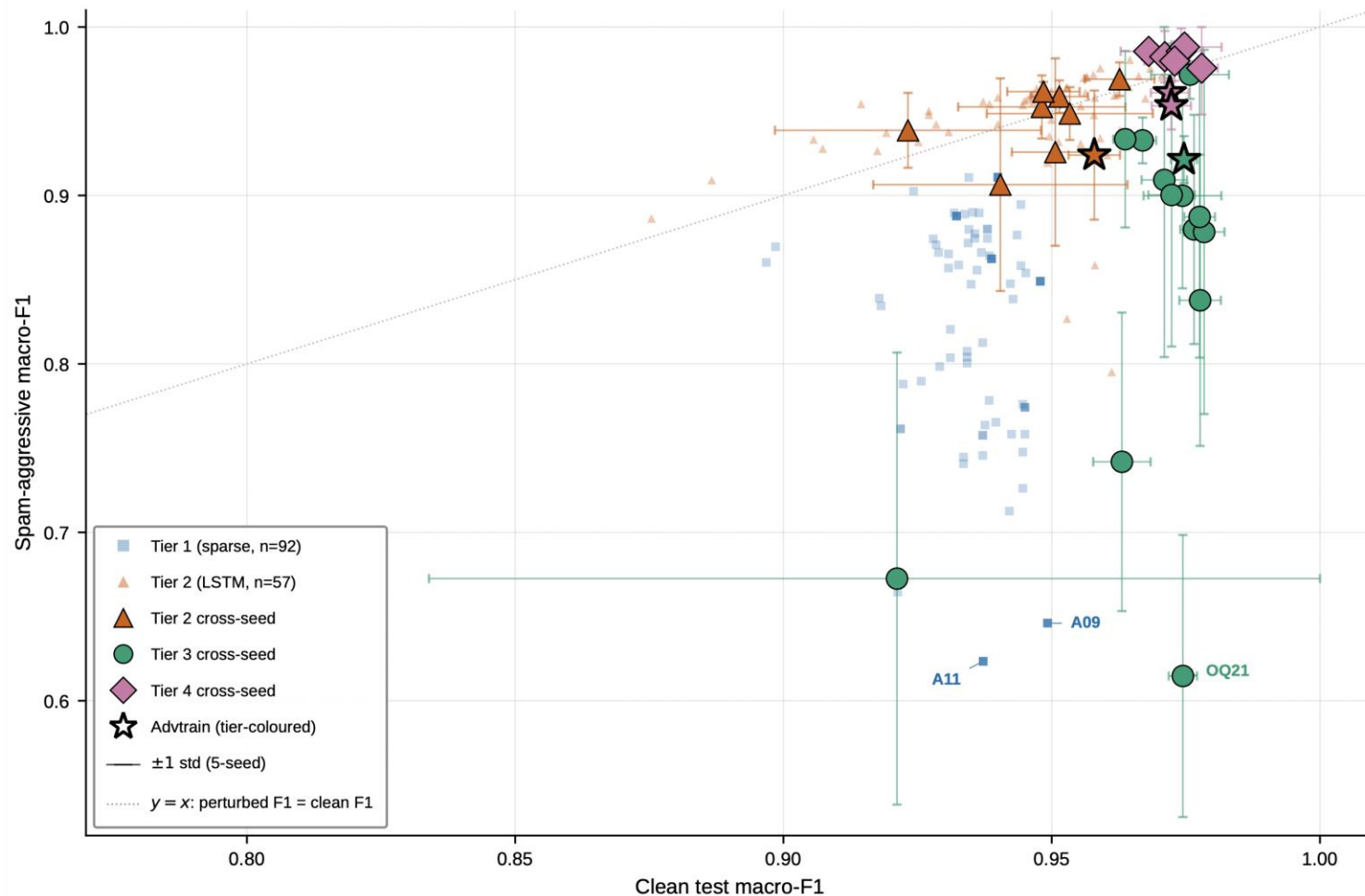


- Clean F1 alone favours clean strong models
- Joint full ranking favours robust models
- Adversarial training becomes important when ham side diagnostic is included

Five-panel deployment-vs-full leaderboards (37 cross-seed families): clean / spam-adv / ham-adv / joint-deploy / joint-full (all bars: 5-seed cross-seed mean)

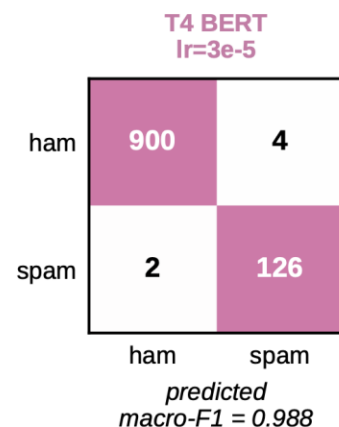
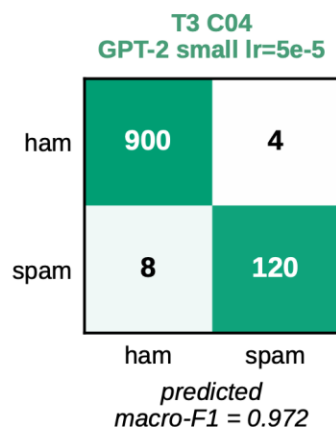
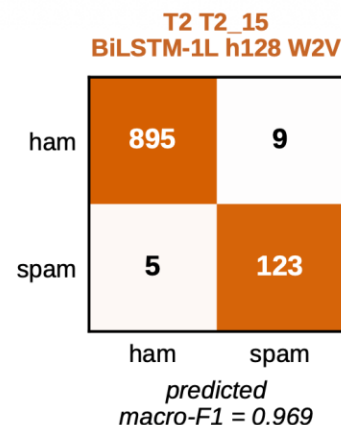
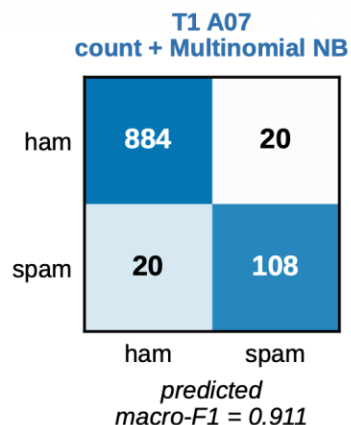


Spam side robustness



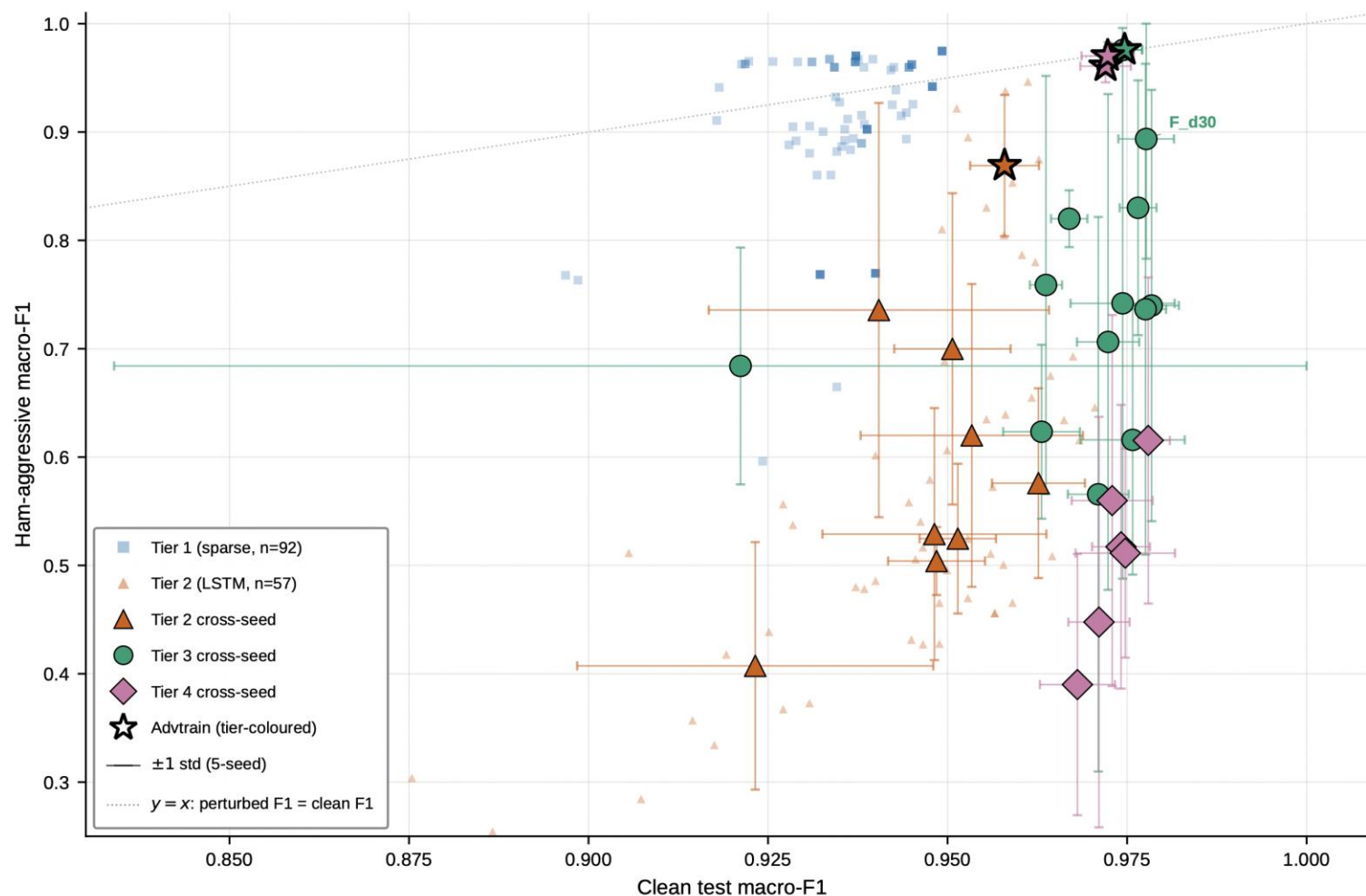
- Spam side perturbation models realistic obfuscation
- Most models stay close to clean performance
- Median spam side delta is around 0.03
- Main fragile exceptions: A09, A11 and OQ21
- Overall, realistic spam obfuscation is mostly handled well

Spam side: most models still detect spam



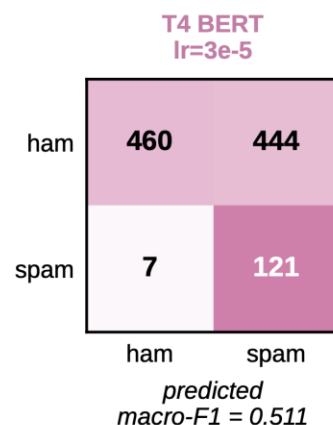
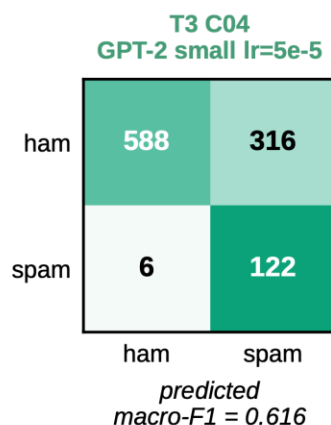
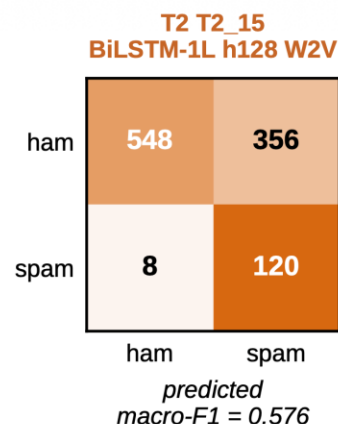
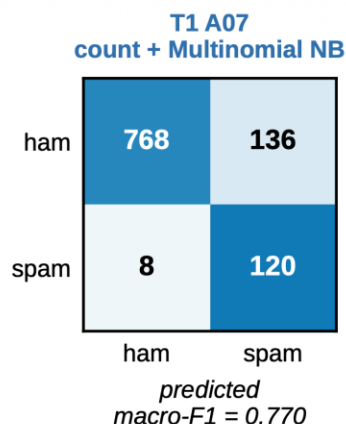
- Same realistic spam side test
- All four tier winners keep strong performance
- BERT misses only 2 of 128 spam messages

Ham side diagnostic and OOV bias



- Ham side perturbation gives a very different result
- 26 of 33 baseline families are ham fragile
- 888 of 904 ham rows become harder under aggressive perturbation
- Clean strong models can still collapse on perturbed ham
- This reveals hidden OOV bias

Ham side: the hidden collapse is visible

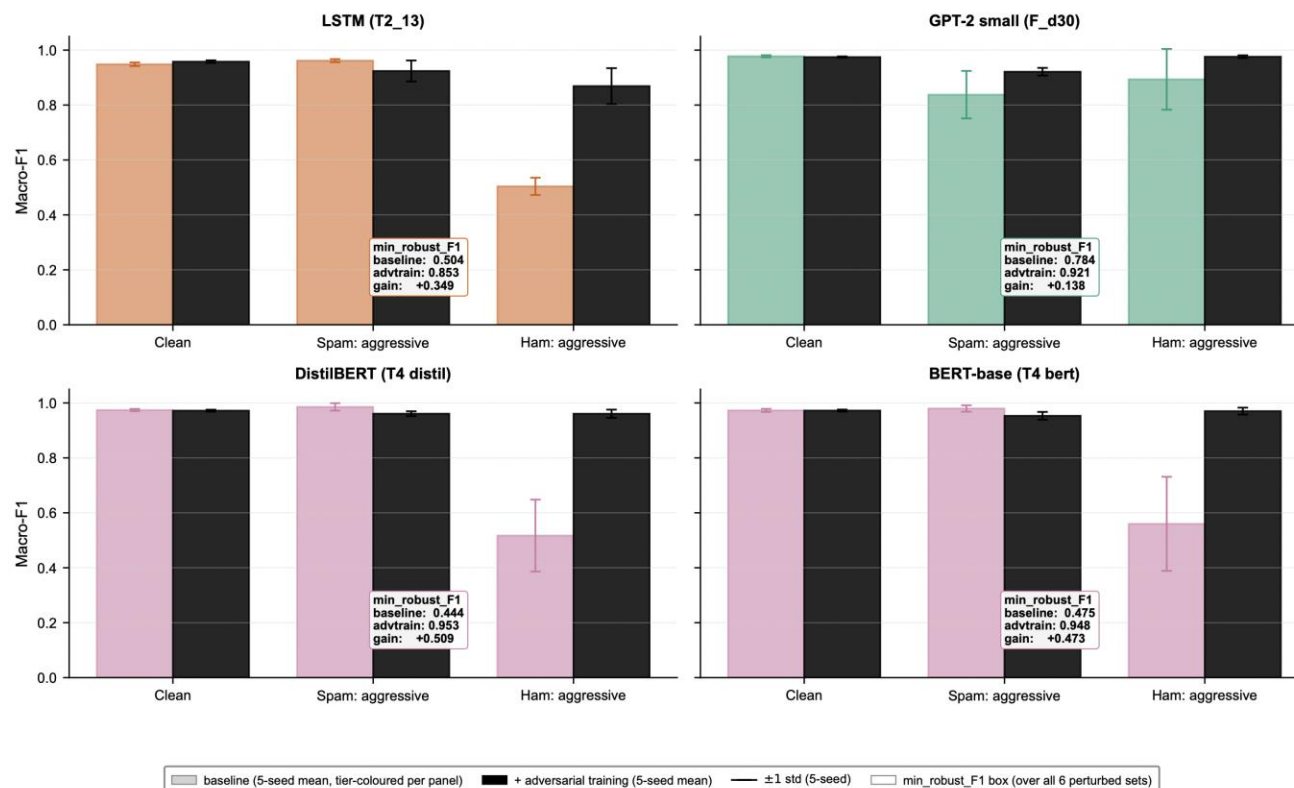


- The same winners as the spam side slide
- Ham misclassification rises strongly
- BERT misclassifies about 49 percent of ham as spam

Mitigation and conclusion



Adversarial training generalises across four trained model families, but residual ham-side gap depends on architecture



- Adversarial training strongly reduces ham side fragility
- Transformer models close the gap almost completely
- LSTM improves, but keeps some residual fragility
- Clean F1 cost is small
- Final lesson: robustness aware evaluation changes model selection

Appendix:

Cross-seed family decoder for the 37 families some of which are used throughout this presentation. Description summarises the architecture and key distinguishing hyperparameters; the full per-cell hyperparameter records are in each tier's reports/metadata/ folder. Adv? marks the four adversarial-training families. T2 w2v-corpus-100 denotes a Word2Vec model trained on the SMS corpus itself (in-domain, 100-dim).

Family ID	Description	Adv?	Family ID	Description	Adv?
<i>Tier 1: sparse vectorizers (n=6)</i>			<i>Tier 3: GPT-2 from scratch (n=14)</i>		
A09	count+lr+light (ngram 1-2)	no	A06_B5	gpt2-medium, last_block	no
B02	tfidf+lr+light (ngram 1-1)	no	A08_B2	gpt2-xl, last_block	no
A05	tfidf+svc+light (ngram 1-2)	no	A09_B3	gpt2-xl, all-layers	no
A11	count+svc+light (ngram 1-2)	no	A12_B4	gpt2-small, no_padding	no
A07	count+nb+light (ngram 1-2)	no	A17_B1	gpt2-small, averaged	no
A01	tfidf+nb+light (ngram 1-2)	no	F_d00	gpt2-small, dropout=0.00	no
<i>Tier 2: recurrent (n=9)</i>			F_d05	gpt2-small, dropout=0.05	no
T2_04	LSTM-bi-h128 (1L), glove-300	no	F_d20	gpt2-small, dropout=0.20	no
T2_05	LSTM-uni-h128 (1L), w2v-corpus-100	no	F_d30	gpt2-small, dropout=0.30	no
T2_06	LSTM-bi-h128 (1L), w2v-corpus-100	no	F_d30_advtrain	gpt2-small, advtrain (d=0.30)	yes
T2_09	LSTM-bi-h64 (1L), w2v-corpus-100	no	E03	gpt2-medium, last_two_blocks	no
T2_11	LSTM-bi-h128 (1L), w2v-corpus-100 (alt)	no	C01	gpt2-small, last_two_blocks	no
T2_13	LSTM-bi-h128 (2L), w2v-corpus-100	no	C04	gpt2-small, flexible_token (all)	no
T2_15	LSTM-bi-h128 (1L), w2v-corpus-100 (alt2)	no	OQ21	gpt2-xl, no_padding	no
T2_19	LSTM-bi-h128 (1L), random-300	no	<i>Tier 4: BERT/DistilBERT (n=8)</i>		
T2_13_advtrain	LSTM-bi-h128 (2L), advtrain	yes	T4_distil_lr2e-5	DistilBERT (lr=2e-5)	no
			T4_distil_lr3e-5	DistilBERT (lr=3e-5)	no
			T4_distil_lr5e-5	DistilBERT (lr=5e-5)	no
			T4_bert_lr2e-5	BERT-base (lr=2e-5)	no
			T4_bert_lr3e-5	BERT-base (lr=3e-5)	no
			T4_bert_lr5e-5	BERT-base (lr=5e-5)	no
			T4C_advtrain_distil_lr5e-05	DistilBERT, advtrain	yes
			T4C_advtrain_bert_lr5e-05	BERT-base, advtrain	yes